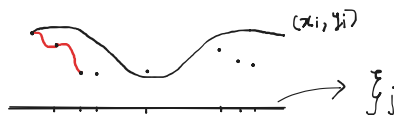


Cubic Splines

We fix $\zeta_1, \dots, \zeta_k$ (not necessarily related to) $x_1, \dots, x_n$

$$x_{(1)} < \dots < x_{(n)}$$

Do not want ζ 's to be far from $(x_{(1)}, x_{(n)})$



We require that f satisfies

- (i) f is a cubic polynomial between $(\zeta_j, f(\zeta_j))$ and $(\zeta_{j+1}, f(\zeta_{j+1}))$
- (ii) f is continuous at the knots
- (ii) f has continuous first and second derivative at each knot.

Instead of cubic, in general, splines can be of any specified degree.

Among all f 's satisfying above, choose the one that minimizes

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

Result: we can write f in terms of $k + 3$ basis for

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 h(x - \zeta_1) + \beta_5 h(x - \zeta_2) + \dots + \beta_{k+3} h(x - \zeta_k)$$

$$h(u) = u^3 I(u > 0)$$

How to choose k and $\zeta_1, \dots, \zeta_k$

ζ_i 's are the quantiles.

k is chosen by cross-validation (cv)

Spline which is linear for $x < \zeta_{(1)}$ and $x > \zeta_{(k)}$ but cubic in between is called a **natural cubic spline**.

Smoothing Splines:

$$\underbrace{\sum (y_i - f(x_i))^2}_{\text{goodness of fit}} + \lambda \underbrace{\int_{x_{(1)}}^{x_{(n)}} |f''(x)|^2 dx}_{\text{penalty for roughness}}$$

λ provides a balance between the 2 goals of fit vs smoothness. Called smoothness penalty/parameter. Chosen λ by cross-validation.

result: Solution on smoothing spline is the natural cubic spline.